Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Conversational AI, Agentic AI, Prompt Engineering, Model Context Protocol (MCP), Self-Serve Analytics, Automated QA Testing, Data Governance & Confidentiality, Dimensional Data Modeling, Star/Snowflake Schemas, SCD Type 1/2, Predictive Modeling, Time-Series Forecasting, Data Quality Auditing, Root Cause Analysis (RCA)

Programming Languages & Tools: GitHub Copilot, KIRO IDE, Azure Data Factory (ADF), Azure Synapse, Azure SQL, Databricks, Snowflake, Apache Spark, PySpark, Python, Scikit-Learn, XGBoost, Power BI, Tableau, Airflow, One Automation

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Data Analyst & AI Tools Builder, People & Operational Analytics — **May 2022 – Present**

- **Engineered advanced conversational AI and automated testing frameworks** utilizing SpectrumIQ to build QA testing systems that cut manual evaluation turnaround by 50% while embedding analytical methodologies into self-serve AI tools.
- **Leveraged Agentic AI and prompt engineering via KIRO IDE** for automated data profiling, regex classification, and building modular AI capabilities, accelerating data tool delivery and capability expansion by 40%.
- **Built automated data quality audit frameworks and firewalls** handling high-throughput data processing to ensure strict data governance, confidentiality, and accuracy when scaling sensitive operational metrics.
- **Developed high-performance dimensional data models and SQL pipelines** using Star/Snowflake schemas and SCD Type 1/2 to ensure governed, trustworthy, self-serve data answers across enterprise reporting workflows.
- **Partnered with cross-functional teams to monitor production batch pipelines** via Airflow and One Automation, maintaining 99.9% data availability SLAs and resolving pipeline bottlenecks through rigorous Root Cause Analysis (RCA).

Graduate Data Science Intern & Analytics Researcher — **May 2022 – Present**

- **Implemented robust machine learning predictive models** using XGBoost, Random Forests, and Scikit-Learn to analyze customer behavior patterns, improving prediction reliability and tooling accuracy by 20%.
- **Applied advanced SARIMA time-series forecasting algorithms** to optimize workforce capacity planning and resource allocation, successfully reducing operational response times by 20%.
- **Designed executive Key Operating Metric (KOM) dashboards** in Power BI (DAX, measures) and Tableau to translate complex analytical datasets into intuitive, self-serve visual decision-making tools.

Petrochemical Engineering Graduate — **May 2022 – Present**

- **Applied rigorous mathematical modeling and statistical analysis** to complex multivariate process control engineering datasets, laying the foundational framework for advanced data analytics and predictive modeling.
- **Optimized multi-variable industrial systems and operational workflows** through systematic troubleshooting, structured root-cause investigation, and data-driven capacity planning methodologies.
- **Spearheaded technical documentation and cross-functional reporting** to bridge complex analytical insights with actionable operational strategies for diverse stakeholder groups.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.